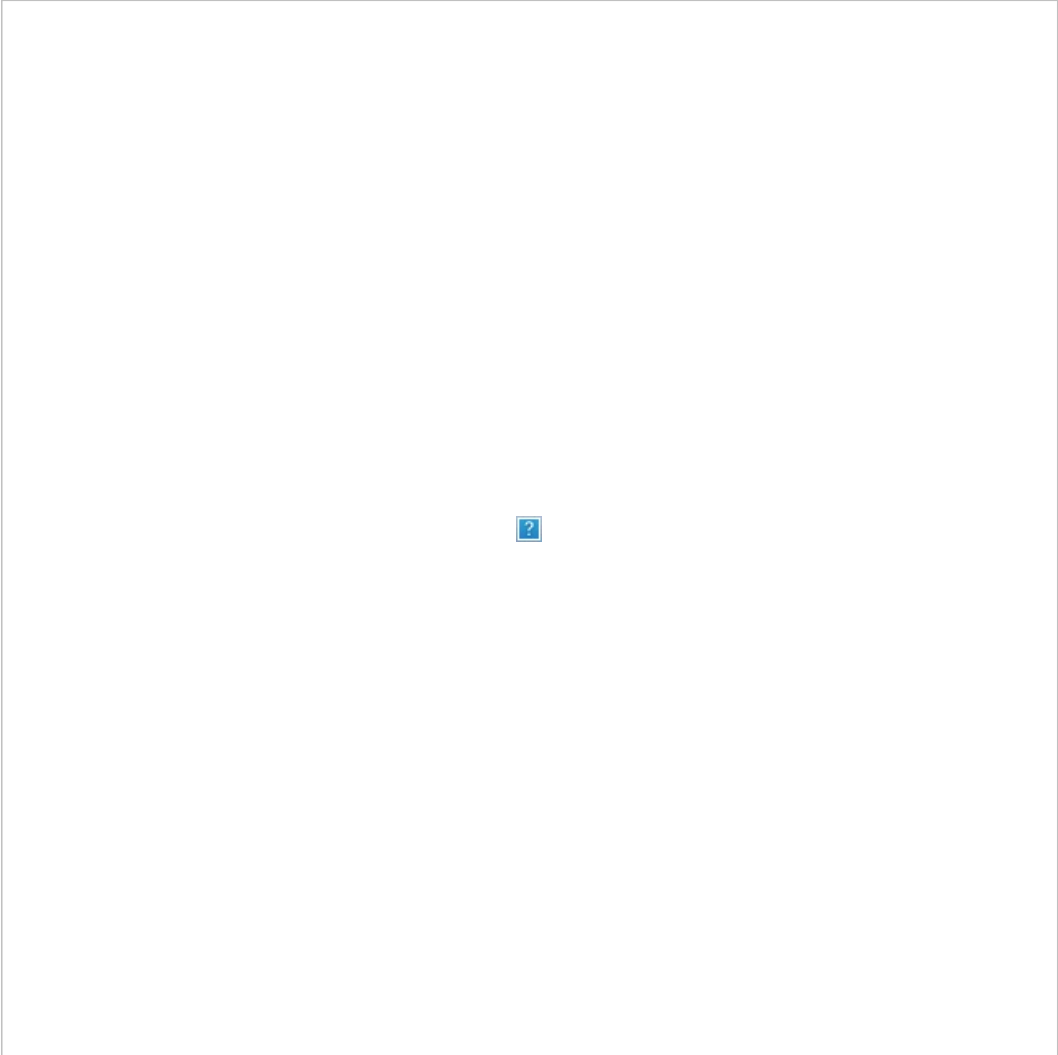




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分布式追踪技术综述

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Survey of State-of-the-art Distributed Tracing Technology

Author:

Affiliation:

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摘要:

随着分布式软件系统在各个行业的广泛应用，如何提升系统运维效率，保障其服务的可靠与稳定，得到了学术界与工业界的关注。分布式软件系统其规模庞大、结构复杂、持续更新且大量服务请求并发执行的特点，给分布式软件系统的运维任务带来了严峻的挑战。传统的以组件/节点/进程/线程为中心的系统监控与追踪方法难以支持分布式软件的故障诊断、性能调优、系统理解等运维任务。分布式追踪技术识别并提取出分布式软件系统因处理单个服务请求所产生的因果相关的事件，以服务请求为中心对分布式软件系统的行为进行精准、细粒度地刻画，对提高分布式软件系统的运维效率有重要意义。对分布式追踪技术的研究与应用进行了综述，从追踪数据获取、请求事件提取、因果关系判断及请求路径表示这4个方面总结了分布式追踪技术的现状；同时以基于请求执行路径的故障诊断和性能分析为例，讨论了学术界对分布式追踪技术的应用研究；最后，对分布式追踪技术的数据读写依赖问题、通用性问题和评价问题进行了探讨并对未来的研究方向进行了展望。

Abstract:

As distributed computing and distributed systems are being widely applied in various areas, how to improve the efficiency of system operations to guarantee the stability and reliability of the services provided by these distributed systems have gained massive momentum from both academia and industry. However, system operation tasks are confronted with tough challenges due the large scale, the intricate structures and dependency, the continuous updating and concurrent service requests of distributed systems. Previous component-/node-/process-/thread-centric monitoring and tracing methods are not sufficient to support the system operation tasks such as fault diagnosis, performance optimization, and system understanding in a distributed system. To address this issue, distributed tracing is proposed and designed. Distributed tracing identifies all the events belonging to the same request and causally correlates these events. Distributed tracing technology precisely and fine-grainedly depicts the behavior of a distributed system in a service-request or workflow-centric way, which is critical to improve the efficiency of system operations. This paper presents a comprehensive survey of existing research work and application of distributed tracing technology. A research framework is proposed and existing research achievements in this field are compared and analyzed with this framework from four perspectives which are acquiring tracing data, identifying the events from the same request, determining the causal relationships among these events, and representing the request execution path. Then the research work of applying distributed tracing technology to system operation tasks such as fault diagnosis and performance optimization is briefly introduced. Finally, the data dependency issue, the generality issue, and evaluation metrics issue of distributed tracing are discussed and a perspective of the future research direction in distributed tracing technology is presented.

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